

Lecture #33-36

Content to be covered

- ❖ Definition of quantum numbers
- ❖ Orbital and spin angular momentum
- ❖ Space quantization
- ❖ Spin-orbit coupling
- ❖ Designation of states
- ❖ Atoms in magnetic field
- ❖ Normal Zeeman effect

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser

Atomic Structure

- Light emitted from hot body on passing through a prism gives continuous spectrum.
- After passing through a cool gas some wavelength were missing in the spectrum of same light.
- Light emitted from a hot gas, bright lines of certain wavelength were observed through prism.
- Excited atoms emit light of only certain wavelengths.
- The wavelengths of emitted light depend on the element.

Hydrogen Atom Spectra

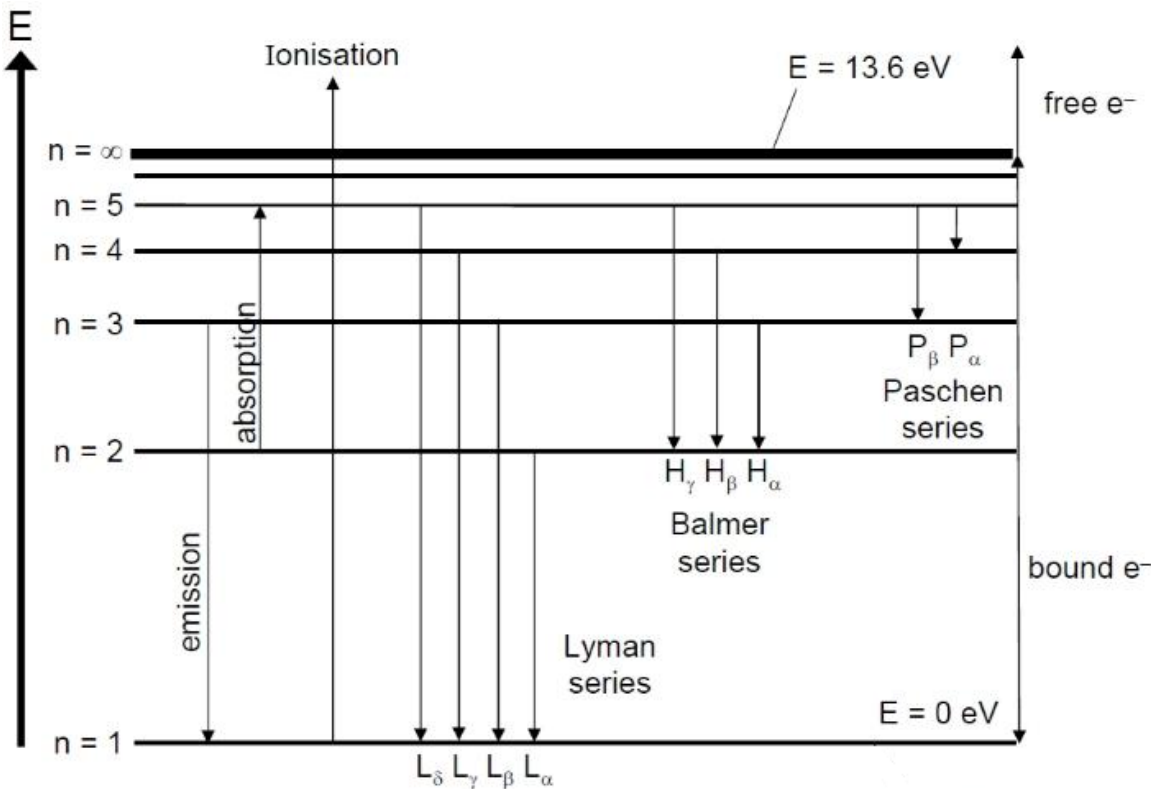
Hydrogen Absorption Spectrum



Hydrogen Emission Spectrum



- ❑ Absorption spectra have black lines (gaps) on a bright background, whereas emission spectra have bright lines on a black background.
- ❑ In both types of spectra the lines are in the same places and also the lines are more closely spaced toward the blue/violet end of the spectrum.



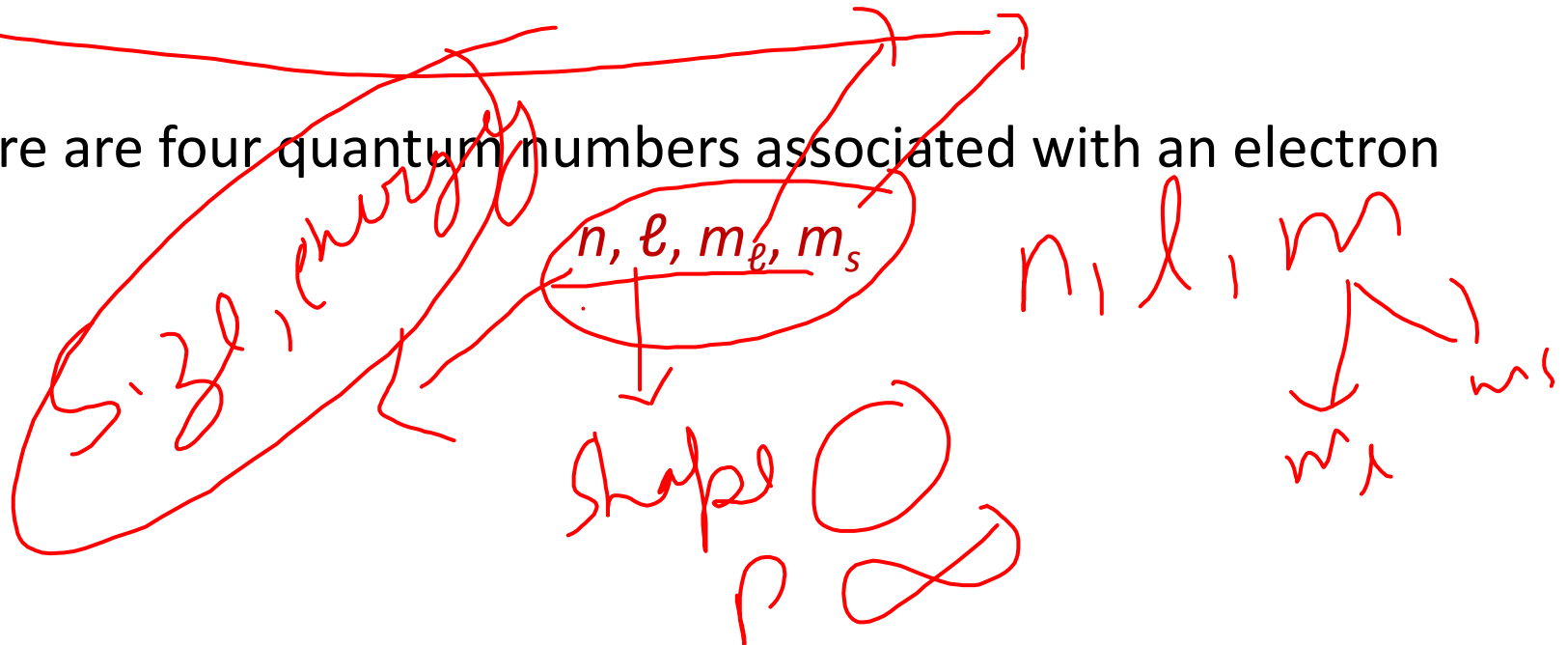
When an atomic gas or vapor is excited, the emitted radiation has a spectrum which contains certain specific wavelength only.

Classical physics can't explain this

Quantum Numbers

A brief overview of the **relationship between the quantum numbers** that come out of the quantum mechanical calculations and the atomic orbital that used to describe the energies and probabilities of finding the electrons in an atom.

There are four quantum numbers associated with an electron

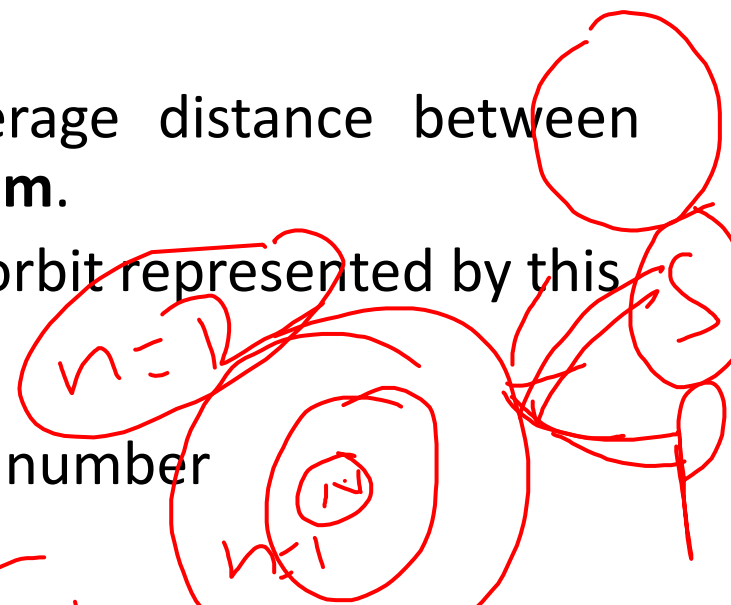


Principle Quantum Number (n): (By Bohr)

It determines the energy and the average distance between electron and nucleus, *i.e.* **size of the atom**.

The maximum number of electrons in an orbit represented by this quantum number is $2n^2$.

$n \rightarrow$ shell $\rightarrow$ K, L, M.... Allowed states : Any number



Azimuthal Quantum Number (ℓ) (by Sommerfield)

Azimuthal or orbital or angular quantum number.

It determines the number of sub shells or sublevels to which the electron belongs. It tells about the **shape of sub-shells**.

$\ell \rightarrow$ sub-shell $\rightarrow$ s, p, d, f.. Allowed states: $\ell = 0$ to $(n-1)$

$n = 2$
 $\ell = 0$ to $(2-1)$
 $\ell = 0$ to 1

$\ell = 0, 1$

Space Quantization

- The **orbital or angular momentum quantum number ℓ** tells us the magnitude of the **orbital angular momentum 'L'**. Such that $L^2 = \ell(\ell+1)\hbar^2 \Rightarrow L = \sqrt{\ell(\ell+1)}\hbar$
- To describe it fully, the direction of this momentum is required.
- We know that when the electron revolves around the nucleus gives rise to current loop and a magnetic field is associated with it.
- Hence, atomic electron possessing an **angular momentum** interacts with this magnetic field.
- The **magnetic quantum number m_ℓ** specifies the direction of 'L' by determining the component of 'L' in the field direction. This phenomenon is called **SPACE QUANTIZATION**.
- Direction of L is perpendicular to the plane of circle.

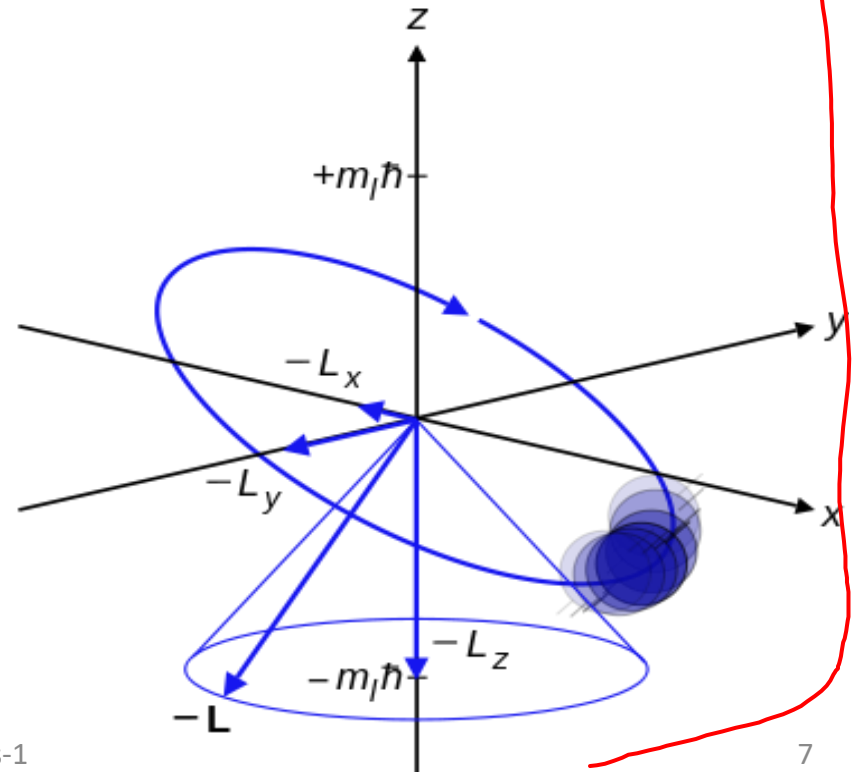
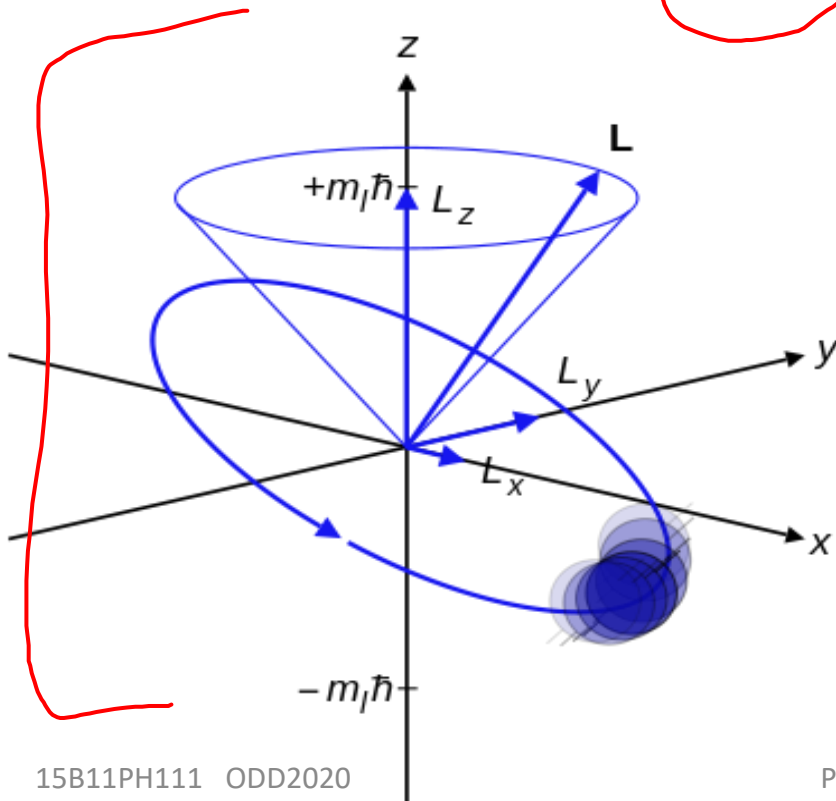
Space Quantization

- Let the direction of magnetic field be parallel to z-axis, then

$$L_z = m_\ell \hbar$$

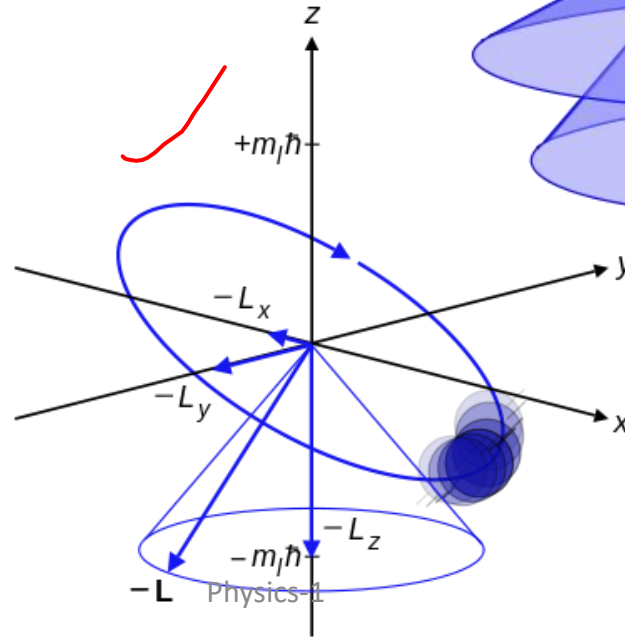
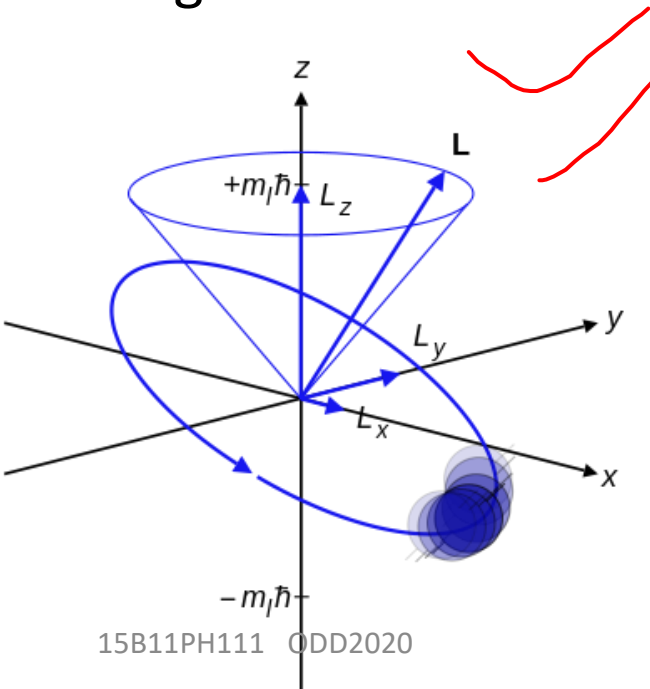
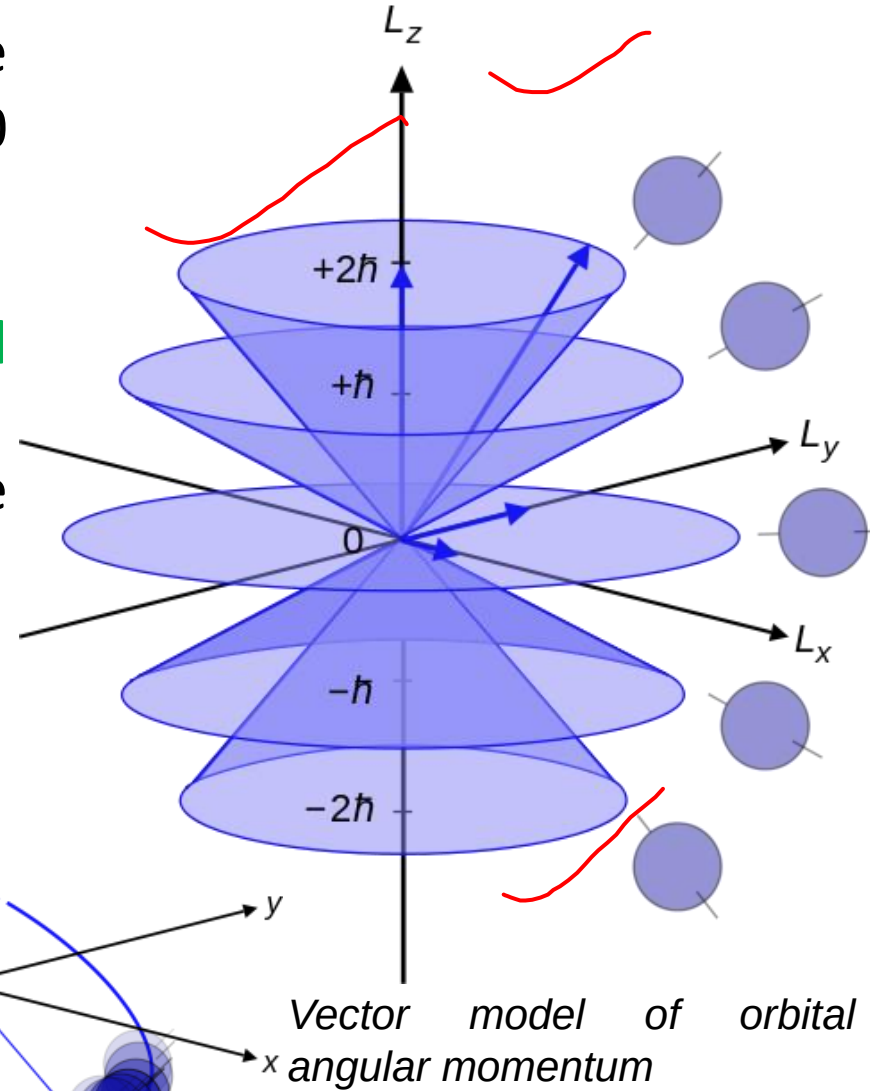
$$m_\ell = -\ell, (-\ell+1), \dots, 0, \dots, (\ell-1), \ell$$

- m_ℓ : The magnetic quantum number.
- possible values of $m_\ell = 2\ell+1$ from $+\ell$ through 0 to $-\ell$.



Space Quantization

- If $\ell = 0$, L_z can have only single value of 0. If $\ell = 1$, L_z has three values $\hbar$, 0 and $-\hbar$.
- It was proposed by Zeeman.
- It gives the number of permitted orientation of sub-shells.
- The space quantization of a molecule having $\ell = 2$:



Spin Quantum Number : s

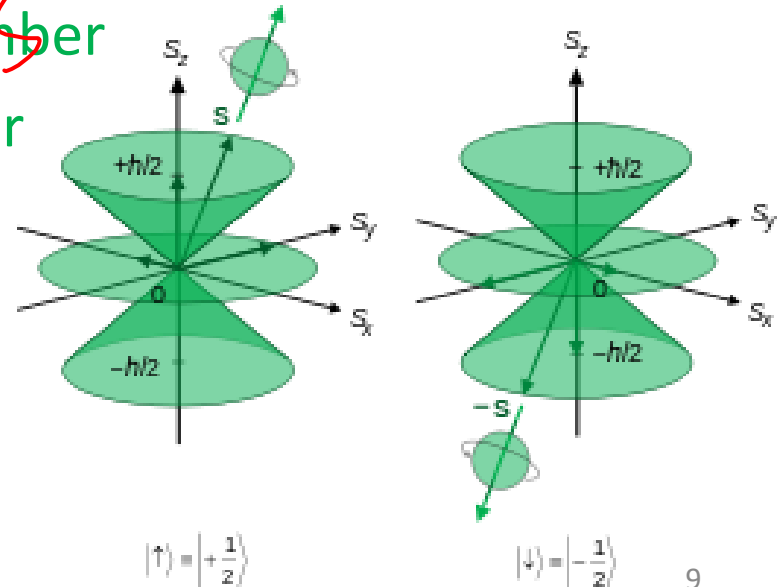
- Each electron has an intrinsic angular momentum which is same for all electron.
- The spin quantum number is defined as $s = 1/2$
- Spin angular momentum $S^2 = s(s+1)\hbar^2$
- The spin motion of the electron around its own axis also produces a magnetic field

n, l, m

$$S_z = m_s \hbar$$

$m_s \rightarrow$ The spin projection quantum number
or magnetic quantum number for
spin motion

$$m_s = \pm 1/2$$



Total Angular Momentum : J

When the orbital angular momentum (ℓ) and spin angular momentum (s) are coupled, the total angular momentum is of the general form for quantized angular momentum

$$J^2 = j(j+1)\hbar^2$$

where, j = total angular momentum quantum number.
such that

$$j = (\ell \pm s) = (\ell \pm 1/2)$$

This gives a z-component of angular momentum

$$J_z = m_j \hbar, \quad m_j = -j, (-j+1), \dots, (j-1), j$$

Total Angular Momentum : J

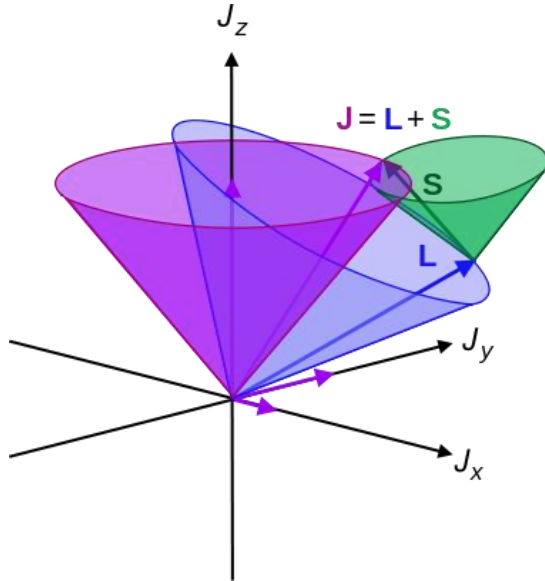
- As long as external interactions are **not** extremely strong, the total angular momentum of an electron can be considered to be conserved and j is said to be a "good quantum number".
- This quantum number is used to characterize the splitting of atomic energy levels, such as the spin-orbit splitting which leads to the sodium doublet.
- Since J_z , S_z and L_z are scalar quantities so

$$J_z = L_z \pm S_z$$

$$m_j = m_\ell \pm m_s$$

- since m_ℓ is an integer and m_s is $\frac{1}{2}$; so m_j must be half-integer.

L-S Coupling



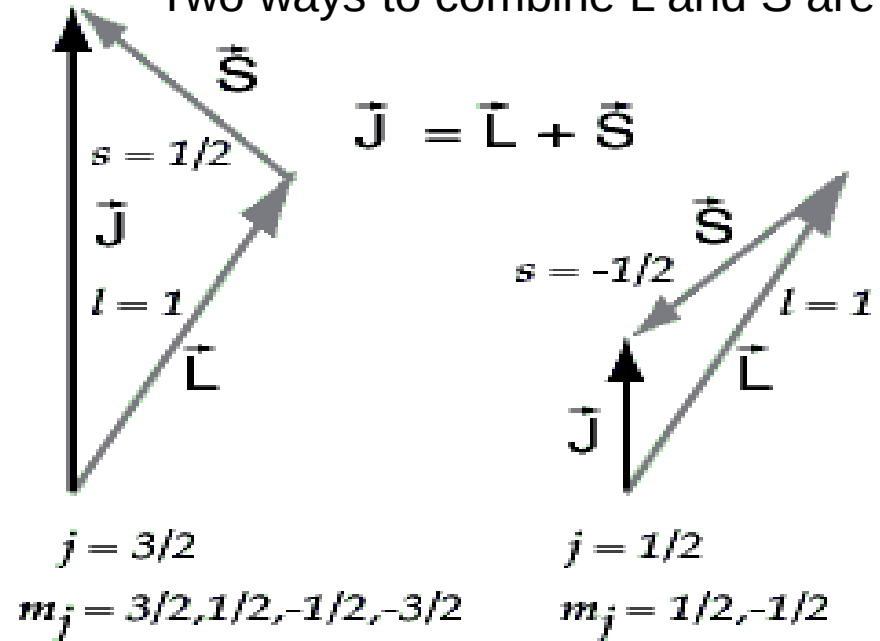
This phenomenon is called L-S coupling.

"Vector cones" of total angular momentum $\mathbf{J}$ (purple), orbital $\mathbf{L}$ (blue), and spin $\mathbf{S}$ (green). The cones arise due to quantum uncertainty between measuring angular momentum components

Pauli exclusion principle

No two electrons can have all their quantum numbers alike.

Two ways to combine L and S are



$$j = |l \pm s|$$

$$|l + s| \rightarrow |l - s|$$

$$l = 1, s = \frac{1}{2} \rightarrow j = \frac{1}{2}, \frac{3}{2}$$

Coupling of orbital momenta

Consider two electrons with l_1 & l_2

$$L = |l_1 \pm l_2| \Rightarrow |l_1 - l_2| \rightarrow (l_1 + l_2)$$

e.g. if $l_1 = 1$ & $l_2 = 2$; then, $L = 0(S), 1(P), 2(D)$

Spin Multiplicity

Indicates the number of possible quantum states of a system with given principal spin quantum number S .

The different states along any direction are distinguished by the spin quantum number S , which can take the values $-S, (-S+1), \dots, (S-1), S$.

$m_s = 2S+1$, where S is the number of singly occupied electrons multiplied by the Electron Spin Quantum Number, m_s .

Spin Multiplicity

For $S=0 \rightarrow$ singlet state.

For $S=1/2 \rightarrow$ doublet. ✓ $M = 2$

For $S=1 \rightarrow$ triplet, and so on. $M = 3$

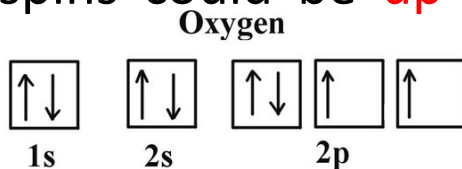
The most important application is two electrons.

A single free electron has $S=1/2$; it is therefore always in a doublet state.

Two electrons can pair up in a *singlet or in a triplet state*. Normally the singlet is the ground state.

E.g. oxygen has two singly occupied electrons which could have spin multiplicity of 3.

This means that the spins could be up-up or up-down or down down, total 3 possibilities.



Using the formula, Spin Multiplicity of oxygen = $2S+1 = 2(2 \times 1/2)+1 = 3$, where $S =$ two singly occupied electrons $\times m_s$ (m_s always equal to $1/2$).

Designation of states: For two electron case

Coupling of orbital angular momenta

- $\ell_1 \rightarrow$ orbital quantum number for first electron
- $\ell_2 \rightarrow$ orbital quantum number for second electron

After coupling resultant orbital quantum number:

$$\mathbf{L} = |\ell_1 - \ell_2| \text{ to } (\ell_1 + \ell_2)$$

If $\ell_1 = 1$ and $\ell_2 = 1$ then $\mathbf{L} = 0$ to 2 Thus $\mathbf{L} = 0, 1, 2$

Coupling of spin angular momenta:

- $s_1 \rightarrow$ spin quantum number for first electron
- $s_2 \rightarrow$ spin quantum number for second

After coupling resultant spin quantum number:

$$\mathbf{S} = |s_1 - s_2| \text{ to } (s_1 + s_2)$$

If $s_1 = 1/2$ and $s_2 = 1/2$ then $\mathbf{S} = 0, 1$

SPIN ORBIT COUPLING

- $\mathbf{J} = |\mathbf{L}-\mathbf{S}| \text{ To } (\mathbf{L}+\mathbf{S})$

$\mathbf{J}$ is the total inner quantum number

- If $\mathbf{L}=1$ and $\mathbf{S}= 0, 1$
- $\mathbf{L}=1, \mathbf{S}=0 \rightarrow \mathbf{J} = 1$
- $\mathbf{L}=1, \mathbf{S}=1 \rightarrow \mathbf{J} = 0, 1, 2 .$

State Designation: $n^M L_j$

- $n \rightarrow$ Principle quantum number
- $L \rightarrow$ Total Orbital Quantum Number
- $J \rightarrow$ Total Inner Quantum Number
- $M \rightarrow$ Spin Multiplicity = $(2S+1)$, $S \rightarrow$ Total Spin
- IF $S = 0$ $M = 1 \rightarrow$ Singlet State
- IF $S = \frac{1}{2}$ $M = 2 \rightarrow$ Doublet State
- IF $S = 1$ $M = 3 \rightarrow$ Triplet State

Atoms with one electron in the outermost shell will give rise **doublet** in atomic spectrum

Atoms with two electron in the outermost shell will give rise **singlet** and **triplet** in atomic spectrum.

Hydrogen atom

ground state

$n=1, l=0 \rightarrow S$ state, $s=1/2$

$j=1/2$ ($l \pm s$),

$M=2$ (spin multiplicity = $2s+1$)

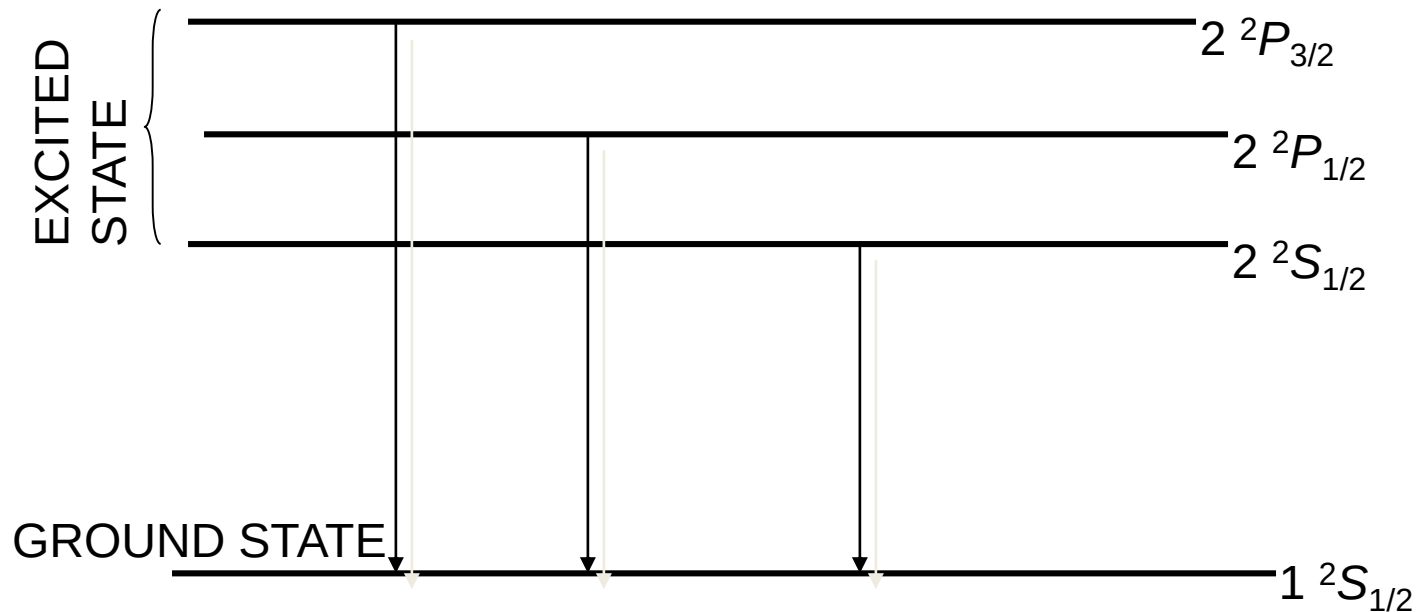
Designation or Term symbol: $1^2S_{1/2}$

first excited state

$n=2, l=0, 1$ as $l=0 \rightarrow (n-1)$ so state is S and P state

1) $l=0, s=1/2, j=1/2, M=2$ so $2^2S_{1/2}$

2) $l=1, s=1/2, j=1/2, 3/2, M=2$ so $2^2P_{1/2}, 2^2P_{3/2}$



WHAT WILL BE THE DESIGNATION FOR $n = 3$ level?

Helium atom

ground state $1s^2$

$l_1 = 0, l_2 = 0$; Hence, $L = 0 \rightarrow S$ state,

$s_1 = 1/2, s_2 = 1/2$; Hence, $S = 0, 1$ { $S=1$ is not possible}

$J=0$ ($L \pm S$),

$M=1$ (spin multiplicity = $2s+1$)

Designation or Term symbol: 1^1S_0

excited state $1s^1 2s^1$

$l_1 = 0, l_2 = 0$; Hence, $L = 0 \rightarrow S$ state

$s_1 = 1/2, s_2 = 1/2$; Hence, $S = 0, 1$

1) $L = 0, S = 0, J = 0; M = 1$ so 2^1S_0

2) $L = 0, S = 1, J = 1, M = 3$ so 2^3S_1

excited state $1s^1 2p^1$

$l_1 = 0, l_2 = 1$; Hence, $L = 1 \rightarrow P$ state

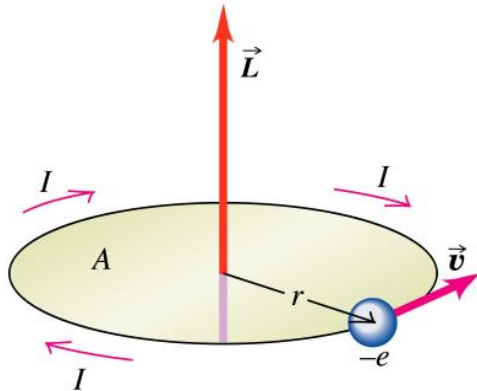
$s_1 = 1/2, s_2 = 1/2$; Hence, $S = 0, 1$

1) $L = 1, S = 0, J = 1; M = 1$ so 2^1P_1 singlet

2) $L = 1, S = 1, J = 0, 1, 2; M = 3$ so $2^3P_0, 2^3P_1, 2^3P_2$
triplet

Zeeman Effect

Atom in magnetic field



Now let us consider motion of an electron in the orbital of a hydrogen atom as a current loop.

$$\text{Magnetic moment } \mu = IA$$

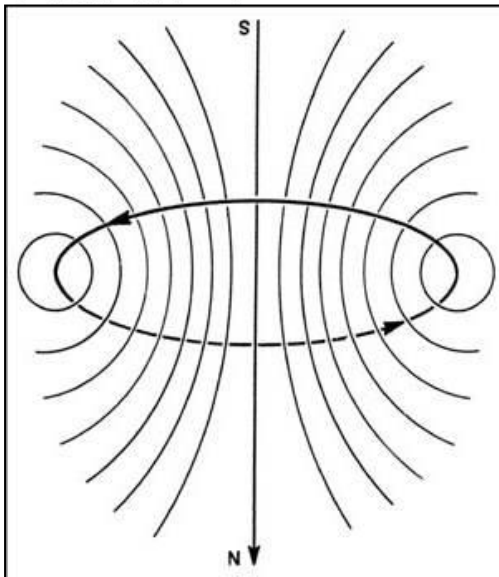
I = current; A = Area the current encloses

Suppose electron's orbital frequency is f

$$I = \frac{q}{t} = -ef \quad \Rightarrow \quad \mu = -ef\pi r^2 \dots \text{Eq. (1)}$$

Angular momentum of the electron

$$L = mvr = m(r\omega)r = mr^2 2\pi f$$
$$\Rightarrow L = 2\pi mfr^2 \dots \text{Eq. (2)}$$



Atom in magnetic field

From Eqs. (1) and (2)

$$\mu = -\left(\frac{e}{2m}\right)L$$

$-(e/2m)$ is referred as gyromagnetic ratio. Negative sign means that μ is in opposite direction to that of L .

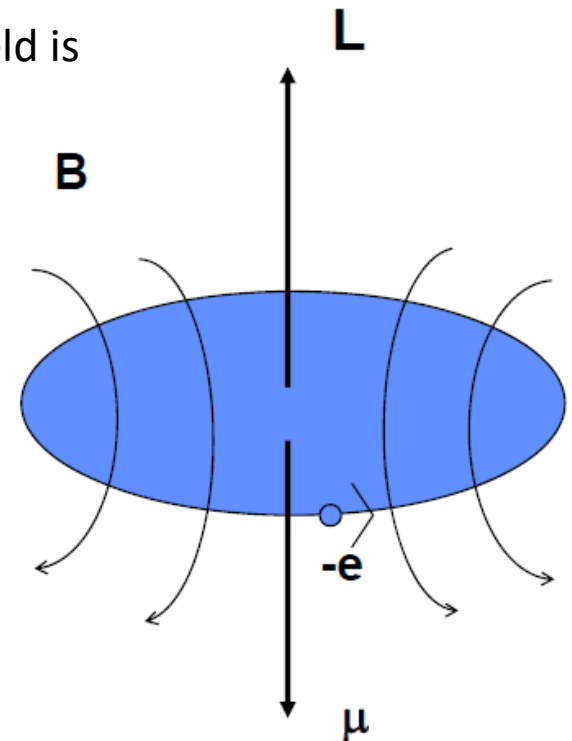
The magnetic potential energy of an atom in a magnetic field is

$$U_m = -\mu B \cos \theta = \left(\frac{e}{2m}\right) L B \cos \theta$$

$$\text{Since, } L_z = L \cos \theta \quad \& \quad \cos \theta = \frac{m_l}{\sqrt{l(l+1)}}$$

$$\text{Hence, } U_m = m_l \left(\frac{e\hbar}{2m}\right) B = m_l \mu_B B$$

$$\text{Here, } \mu_B = \frac{e\hbar}{2m} \text{ is called Bohr magneton.}$$

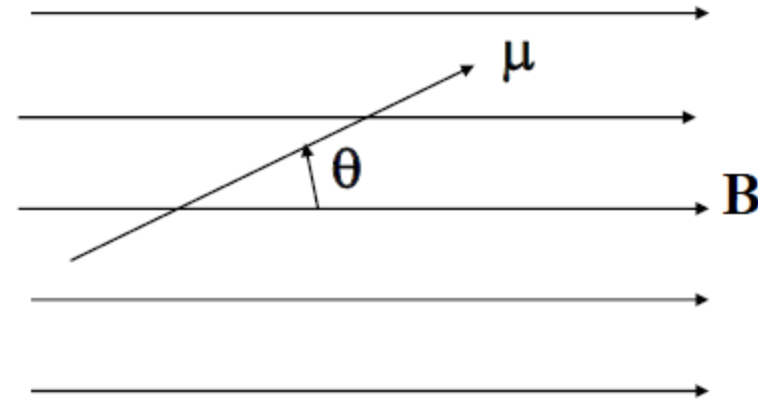


Atom in magnetic field

Consider a magnetic dipole (magnetic dipole moment μ in a magnetic field B . Torque τ on this dipole

$$\tau = \mu B \sin \theta$$

where θ is the angle between μ and B .



$\tau = 0$ when $\theta = 0$ or 180° i. e. dipole is parallel or anti-parallel to the field.

$\tau = \tau_{\max}$ when $\theta = 90^\circ$ i. e. dipole is perpendicular to the field.

Potential energy of dipole inside magnetic field depends on the magnitude of μ of magnetic moment and the orientation of the moment with respect to the field.

$$U_m = \int_{\frac{\pi}{2}}^{\theta} \tau d\theta = \mu B \int_{\frac{\pi}{2}}^{\theta} \sin \theta d\theta \Rightarrow U_m = -\mu B \cos \theta$$

- When μ points in the same direction as B then $U_m = -\mu B$, mean its maximum value.
- Potential energy U_m is zero at $\theta = 90^\circ$ means when μ is perpendicular to B .
- Magnetic dipole tends to align itself with an external magnetic field.

Suppose hydrogen atom is placed in a magnetic field such that its magnetic moment makes an angle θ with the external magnetic field direction, then potential energy would be

$$U_m = -\mu B \cos \theta = -\left(-\frac{e}{2m}\right) L B \cos \theta$$

$$\Rightarrow U_m = \frac{e}{2m} L B \cos \theta$$

$$[\vec{\mu} = -\frac{e}{2m} \vec{L}]$$

$$\Rightarrow U_m = \frac{e}{2m} L_z B$$

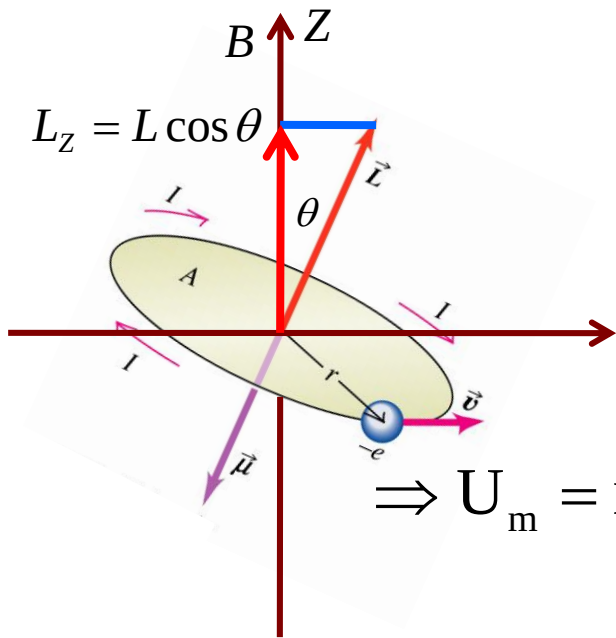
$$\because L_z = m_\ell \hbar$$

$$\Rightarrow U_m = m_\ell \frac{e\hbar}{2m} B$$

$$\Rightarrow U_m = m_\ell \mu_B B$$

$$\mu_B = \frac{e\hbar}{2m} = 9.274 \times 10^{-24} \text{ J/T} = 5.788 \times 10^{-5} \text{ eV/T}$$

Bohr magneton



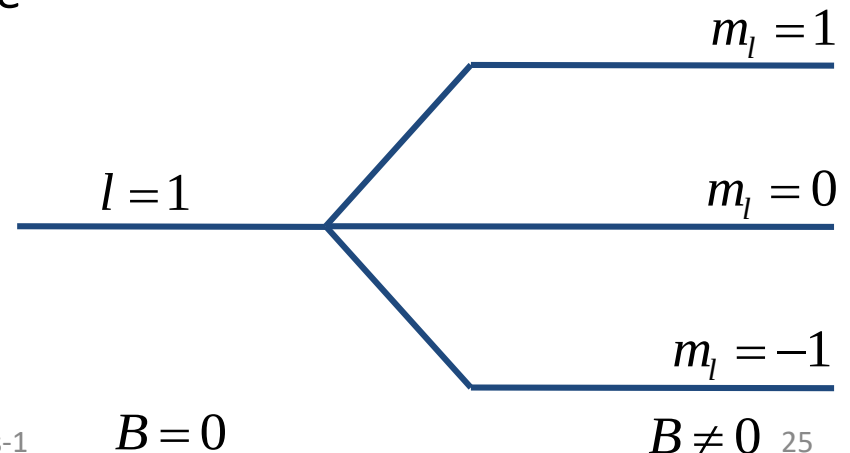
Zeeman Effect

- In magnetic field the energy of a particular atomic state depends on the value of m_l as well as on n .
- A state of total quantum number n breaks up into several sub-states when the atom is in a magnetic field and their energies are slightly more or slightly less than energy of the state in the absence of the field.
- This phenomenon leads to splitting of individual spectral lines into separate lines when atoms radiate in a magnetic field.
- The spacing of the lines depends on the magnitude of the field.
- The splitting of spectral lines by magnetic field is called **Zeeman effect**.
- Changes in m_l are restricted to $\Delta m_l = 0, \pm 1$.
- Normal Zeeman effect consists of splitting of a spectral line of frequency ν_0 into three components whose frequencies are

$$\nu_1 = \nu_0 - \mu_B \frac{B}{h} = \nu_0 - \frac{e}{4\pi m} B$$

$$\nu_2 = \nu_0$$

$$\nu_3 = \nu_0 + \mu_B \frac{B}{h} = \nu_0 + \frac{e}{4\pi m} B$$



Zeeman Splitting for $l = 1$

$$U_m = m_l \mu_B B$$

Consider an atomic electron with orbital quantum number $\ell = 1$.

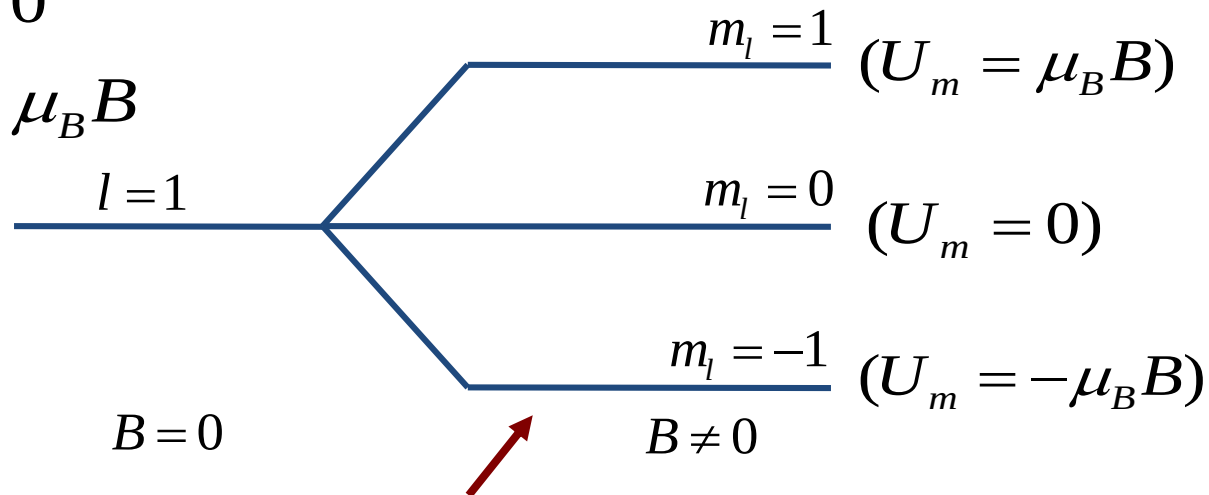
$$\ell = 1 \Rightarrow m_\ell = -1, 0, 1$$

Therefore potential energies of this electron would be

$$U_m = -\mu_B B$$

$$U_m = 0$$

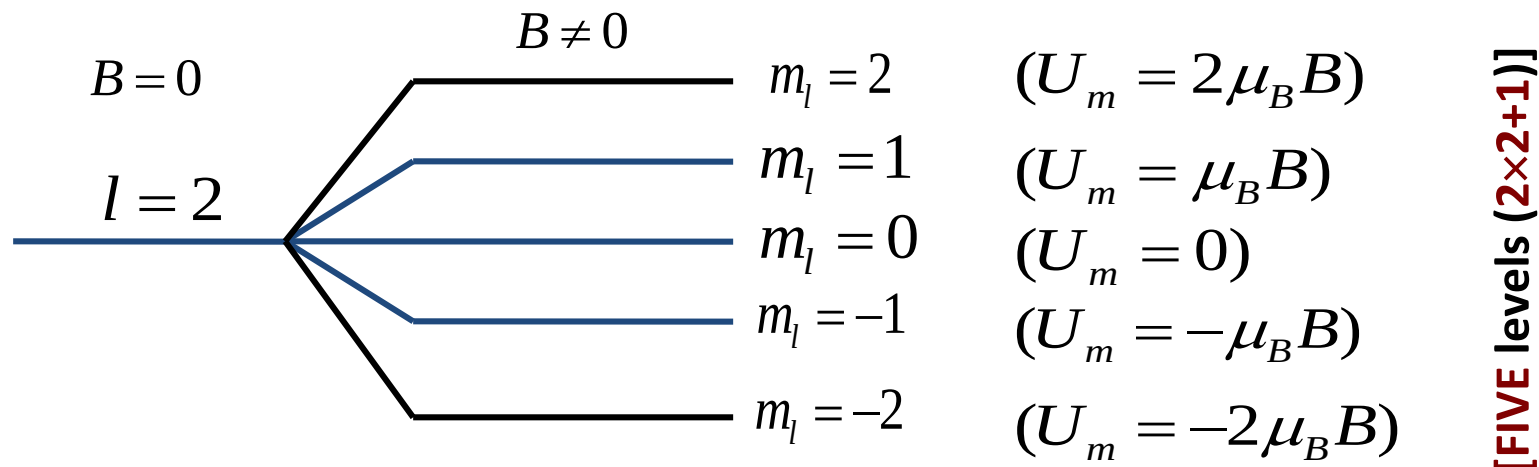
$$U_m = \mu_B B$$



Splitting of energy level: **Zeeman Effect**

Zeeman Splitting for $l = 2$

A state of given orbital quantum number ℓ splits into $2\ell+1$ sub-states which differ in energy by $\mu_B B$. For example, if $\ell=2$ then $m_\ell=0, \pm 1, \pm 2$



Transition between two states of different ℓ is restricted by selection rule given as:

$$\Delta m_\ell = 0, \pm 1$$

Due to this restriction, transition between two states of different ℓ splits into only three components which is called NORMAL ZEEMAN EFFECT.

Zeeman Effect

With magnetic field

No field

$l = 2$

$m_\ell = 2$

$m_\ell = 1$

$m_\ell = 0$

$m_\ell = -1$

$m_\ell = -2$

$h\nu_0$

$h\nu_0 - \frac{e\hbar B}{2m}$

$h\nu_0$

$h\nu_0 + \frac{e\hbar B}{2m}$

$l = 1$

$m_\ell = 1$

$m_\ell = 0$

$m_\ell = -1$

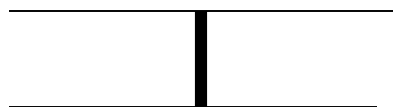
$\Delta m_\ell = -1$

$\Delta m_\ell = 0$

$\Delta m_\ell = 1$

Remember
Selection rule

$$\Delta m_\ell = 0, \pm 1$$



$$\nu_1 = \nu_0 - \frac{e}{4\pi m} B$$

$$\nu_2 = \nu_0$$

$$\nu_3 = \nu_0 + \frac{e}{4\pi m} B$$

Problem

1. A sample of certain element is placed in a 0.300 Tesla magnetic field and suitable excited. How far apart the Zeeman components of the 450 nm spectral line of this element?

Separation of the Zeeman component is given by

$$\Delta \nu = \frac{eB}{4\pi m}$$

$$\nu = \frac{c}{\lambda} \Rightarrow |d\nu| = \frac{c}{\lambda^2} d\lambda$$

$$\text{Hence, } \Delta\lambda = \frac{\lambda^2 \Delta\nu}{c} \text{ or, } \Delta\lambda = \frac{eB\lambda^2}{4\pi mc} = 2.83 \times 10^{-12} \text{ m}$$

2. The Zeeman component of a 500 nm spectral lines are 0.0116 nm apart when magnetic field is 1.00 T. Find the ratio of e/m for the electron from this data.

$$\Delta\lambda = \frac{eB\lambda^2}{4\pi mc} \Rightarrow \frac{e}{m} = \frac{4\pi c \Delta\lambda}{B\lambda^2} = 1.748 \times 10^{11} \text{ C / kg}$$